Exercise 4: Date functions

Question 1

```
SELECT order_id, customer_id, order_date,
       DAYNAME(order_date) AS day_name
from orders
```

order_id	customer_id	order_date	day_name
1001	101	2026-05-01	friday
1002	102	2026-05-02	Saturday
1003	103	2026-05-03	Sunday
1004	104	2026-05-04	monday
1005	105	2026-05-05	Tuesday

Question 2

```
SELECT customer_id, customer_name, signup_date,
       MonthName (signup_date) AS signup_month_name
from customer_signups
```

customer_id	customer_name	signup_date	signup_month_name
201	John	2026-01-15	January
202	Mary	2026-02-20	febrary
203	Peter	2026-03-05	March
204	Sarah	2026-04-18	April
205	Thabo	2026-05-20	May

Question 3

```
SELECT sale_id, product_name, sale_date,
       MONTH(sale_date) AS sale_month
FROM sales;
```

sale_id	product_name	sale_date	sale month
S001	Laptop	2026-01-10	1
S002	Mouse	2026-02-15	2
S003	Keyboard	2026-03-20	3
S004	Monitor	2026-03-25	4
S005	Desk	2026-05-20	5

Question 4

```
SELECT transaction_id, customer_id, transaction_date,
       YEAR(transaction_date) AS transaction_year
FROM transaction;
```

transaction_id	customer_id	transaction_date	transaction_year
T001	101	2024-12-15	2024
T002	102	2025-01-20	2025
T003	103	2025-06-10	2025
T004	104	2026-02-05	2026
T005	105	2026-05-25	2026

Question 5

```
SELECT delivery_id, customer_id, delivery_date,
       DAY(delivery_date) AS day_of_month
FROM deliveries
```

delivery_id	customer_id	delivery_date	day_of_month
D001	101	2026-05-01	1
D002	102	2026-05-08	8
D003	103	2026-05-15	15
D004	104	2026-05-22	22
D005	105	2026-05-29	29

Question 6

```
SELECT employee_id, employee_name, department,
       CURRENT_DATE AS today_date
FROM employees;
```

employee_id	employee_name	department	today_date
301	Mandi	Sales	2026-07-1
302	Brian	IT	2026-07-1
303	Lerato	finance	2026-07-
304	Sipho	HR	2026-07-
305	Alisha	Marketing	2026-07-

Questions 9

SELECT customer_id, customer_name, last_purchase_date,
DATEDIFF (Current_DATE(), last_purchase) AS
days_since_last_purchase
FROM customer_purchase;

customer_id	customer_name	last_purchase_date	day_since_last purchase
501	John	2026-05-01	79
502	Mary	2026-05-10	70
503	Peter	2026-05-15	65
504	Sarah	2026-05-20	60
505	Thabo	2026-05-25	55

Question 10

SELECT order_id, customer_id, order_date,
DATEADD(DAY, 7, order_date) AS expected_delivery_date.

FROM shipping_orders

order_id	customer_id	order_date	expected_delivery_date
6001	101	2026-05-01	2026 05-08
6002	102	2026-05-03	2026-05-10
6003	103	2026-05-05	2026-05-12
6004	104	2026-05-07	2026-05-14
6005	105	2026-05-09	2026-05-16

Question 11

```
SELECT booking_id, customer_id, booking_date,
YEAR(booking_date) AS booking_year
MONTH(booking_date) AS booking_month
DAY(booking_date) AS booking_day
from bookings;
```

booking_id	customer_id	booking_date	booking_year	booking_month	booking_day
B001	101	2026-01-12	2026	1	12
B002	102	2026-02-18	2026	2	18
B003	103	2026-03-22	2026	3	22
B004	104	2026-04-09	2026	4	9
B005	105	2026-05-27	2026	5	27

Question 12

```
SELECT order_id, customer_id, order_date,
       YEAR(order_date) AS order_year, amount
from yearly_orders
WHERE YEAR(order_date) = 2026;
```

Order_id	customer_id	order_date	order_year	amount
7004	104	2026-02-05	2026	1500
7005	105	2026-05-25	2026	2000

Question 15

```
SELECT send_id, customer_id, send_date
DATE_TRUNC ('month', send_date) AS month_end_date
FROM campaign_sends;
```

send_id	customer_id	send_date	month_start_date
C001	101	2026-01-12	2026-01-01
C002	102	2026-02-18	2026-02-01
C003	103	2026-03-22	2026-03-01
C004	104	2026-04-09	2026-04-01
C005	105	2026-05-27	2026-05-01

Question 16

```
SELECT invoice_id, customer_id invoice_date
TO_CHAR (invoice_date, 'month YYYY') AS invoice_month_year
FROM invoice_dates;
```

invoice_id	customer_id	invoice_date	invoice_month_year
INV001	101	2026-01-05	January 2026
INV002	102	2026-02-10	february 2026
INV003	103	2026-03-15	March 2026
INV004	104	2026-04-20	April 2026
INV005	105	2026-05-25	may 2026

Question 17

```
SELECT customer_id, customer_name, date_of_birth,
floor(DATEDIFF(CURRENT_DATE(), date_of_birth)/365.25) AS
customer_age
from customer_birthdays;
```

customer_id	customer_name	date_of_birth	customer_age
901	John	1998-05-10	28
902	mary	1990-06-20	35
903	Peter	2002-03-15	24
904	Sarah	1985-12-01	40
905	Thabo	2000-07-30	25

Question 18

```
SELECT order_id, customer_id, order_date
DAYTIME(order_date) AS day_name,
CASE
WHEN DAYNAME(order_date) IN ('saturday', 'Sunday') THEN
'weekend'
Else 'weekday'
END AS day_type
from weekend_orders;
```

order_id	customer_id	order_date	day_name	data_type
9001	101	2026-05-01	friday	Weekday
9002	102	2026-05-02	Saturday	weekend
9003	103	2026-05-03	Sunday	Weekend
9004	104	2026-05-04	monday	weekday
9005	105	2026-05-05	Tuesday	weekday

Question 19

SELECT transaction_id, customer_id, transaction_date,
QUATER(transaction_date) AS transaction_quarter, amount
FROM quarterly_transactions;

transaction_id	customer_id	transaction_date	transaction_quarter	amount
Q001	101	2026-01-15	1	500
Q002	102	2026-03-20	1	1200
Q003	103	2026-04-10	2	800
Q004	104	2026-07-05	3	1500
Q005	105	2026-10-25	4	2000

Question 20

SELECT order_id, customer_id, order_date,
~~DATEDD~~
DATEDIFF (CURRENT_DATE (), order_date) AS days_since_order,
amount
FROM recent_orders
WHERE DATEDIFF (CURRENT_DATE (), order_date) > 30;

Order_id	customer_id	order_date	days_since_order	amount
R001	101	2026-04-01	109	500
R002	102	2026-04-15	95	1200
R003	103	2026-05-01	79	800
R004	104	2024-05-10	70	1500
R005	105	2026-05-25	55	2000

Bonus - Customer Recency Segment

```
SELECT customer-id
customer_name, last-purchase_date
DATEDIFF ('day', last-purchase_date,
CURRENT_DATE ())
AS day-since-last-purchase
CASE
WHEN DATEDIFF ('day', last-purchase_date,
CURRENT_DATE ()) <=30
THEN 'Active customer'
WHEN DATEDIFF ('day', last-purchase_date
CURRENT_DATE ()) BETWEEN 31 AND 90
THEN 'At Risk customer'
ELSE AS customer-status
from customer_recency;
```

customer_id	customer_name	last-purchase_day	day since last pu	custom status
1001	John	2026-05-25	91	Inactive
1002	Mary	2026-03-10	106	inactive
1003	Peter	2026-04-01	143	inactive
1004	Sarah	2026-02-15	190	inactive
1005	Thabo	2025-12-20	247	inactive